Lorenzo Baladier

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Embedded Software Engineer · Seeking a full-time role in Paris from September 2027

Education

Polytech Nice Sophia – Université Côte d'Azur — Sophia Antipolis, FR
MSc-level Engineering Degree in Computer Science, IoT & CPS, apprenticeship — *2024 – 2027*

IUT Nice Côte d'Azur — Sophia Antipolis, FR
University Diploma in Computer Science (DUT), apprenticeship — *2022 – 2024*

Lycée Pierre et Marie Curie — Menton, FR
French Baccalauréat, Scientific track, EsaBac (French–Italian dual diploma) — *2019 – 2022*

Experience

Embedded & SDV Software Engineer – Apprentice — Sep. 2024 – Present
Ampere Software Technology — *Biot, FR*

- Designed and benchmarked a prototyping platform based on Google's **Software-Defined Vehicle** environment: **CarOS**
- Brought the platform up on **Raspberry Pi** hardware and on **Cuttlefish** virtual devices; set up the **CI/CD** pipelines
- Built end-to-end use cases covering service-oriented communication, vehicle diagnostics and data processing
- Rewrote a Python prototype into a modular **Rust** CLI/TUI tool generating **SOME/IP**, **Protobuf** test commands from GraphQL, unit-tested with trait-based mocks

Robotics & Visual Servoing Intern — May 2026 – Aug. 2026
DNIIT – Da Nang Institute of Innovative Technology — *Da Nang, Vietnam*

- Implemented closed-loop **visual servoing** of an industrial **SCARA** arm for dynamic pick-and-place of moving targets
- Integrated real-time **YOLO** object detection and pose estimation with **ROS 2** and **MoveIt 2**
- Validated kinematics and PID control in **MATLAB** and **Gazebo** before deploying on the physical robot – source code

Software Developer – Apprentice — Sep. 2022 – Jul. 2024
ESI – European Systems Integration — *Le Cannet, FR*

- Developed a **Java** application exporting complex data from a remote database to JSON
- Automated end-to-end tests with Selenium/TestNG, then led their migration to **TypeScript/Cypress**

Projects

CHIP-8 Emulator | *Rust, SDL2* – github.com/Loren844/rust-chip-8 — 2025

- Emulated the CHIP-8 virtual machine: fetch-decode-execute loop, memory, registers, timers, display and keypad

Go-Biking | *C#, SOAP/REST, ActiveMQ* — School project

- Service-oriented route planner combining bike-sharing and routing APIs, with a caching proxy and asynchronous messaging

Technical Skills

Languages: Rust, C, C++, Python, Java, TypeScript, Bash
Embedded & Automotive: Linux, Raspberry Pi, Cuttlefish, Software-Defined Vehicle, SOA, SOME/IP, Protobuf
Tools: Git, GitLab CI/CD, Cypress, Selenium
Spoken Languages: French (native), Italian (fluent, EsaBac), English (TOEIC 955/990)